NOVA IMS INFORMATION MANAGEMENT SCHOOL

Big Data Analysis Project

Spark Usage

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01

Introduction

Main points for usage of Spark

02

Machine Learning

Why Spark is so well regarded for large scale data in Machine Learning

03

Graph Analytics

The difference Spark has in the creation of relationships between data

01. INTRODUCTION

MAIN POINTS FOR SPARK'S USAGE

Modern organizations generate data that is **large in volume**, **high in velocity**, and **diverse in nature**. To **extract value** from this information, companies need tools capable of processing data **at scale**, **in real time**, and across **multiple formats**.

Spark provides a scalable, distributed computing framework designed exactly for these challenges.

It offers a unified ecosystem that includes:

- **ETL and Data Engineering** (RDDs, *DataFrames*, *SparkSQL*)
- **Machine Learning Pipelines** (*MLlib*)
- **Graph Analytics** (*GraphFrames*)
- Seamless integration with external systems, including ***MongoDB***

This allows end-to-end **workflows to run efficiently** on both structured and semi-structured data.

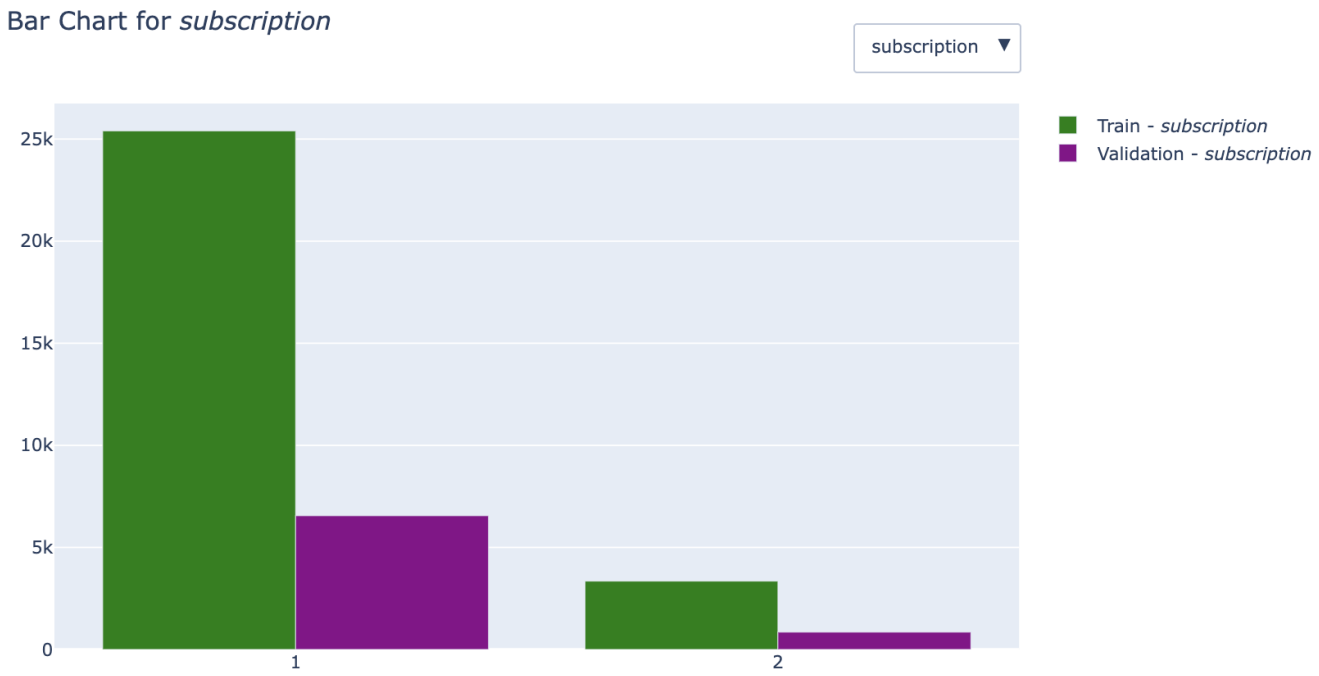
02. MACHINE LEARNING

BANK MAKETING DATASET

SOURCE: Marketing campaigns of a Portuguese bank.

https://www.kaggle.com/datasets/aakashverma8900/portuguese-bank-marketing

GOAL: Predict a binary outcome, whether a client subscribes (2) or not (1) to a term deposit.



CLIENT INFO

- # Age
- Job
- Marital Status
- Education
- Credit default
- # Balance
- Housing
- Personal Loans
- High Effort Clients

CAMPAIGN INFO

- Type of contact
- # Day and month of last contact
- # Duration of last contact
- # Number of contacts in campaign
- Had Previous Contact

02. MACHINE LEARNING

BANK MAKETING DATASET

IMPORTING DATASET

GENERAL CORRECTIONS
Columns' Names
Columns' Datatypes

EXPLORATION ON WHOLE DATASET
Rows (45 211)
Columns (18)

DATA PARTITION (STRATIFIED)
Train | Validation | Test
64% | 16% | 20%

EXPLORATION ON TRAIN & VALIDATION
Shape, Duplicates and Missing Values

VISUALIZATIONS
Target Variable Distribution
Numerical Features
Categorical Features
Features' Relationships

OUTLIERS
Winsorization

FEATURE ENGINEERING
High Effort Client
Had Previous Contact

ENCODING
String Indexer
One Hot Encoder

SCALLING
Normalized

MISSING VALUES
None

FEATURE SELECTION
No Multicollinearity

LOGISTIC REGRESSION

DECISION TREE

RANDOM FOREST

GRADIENT BOOSTING

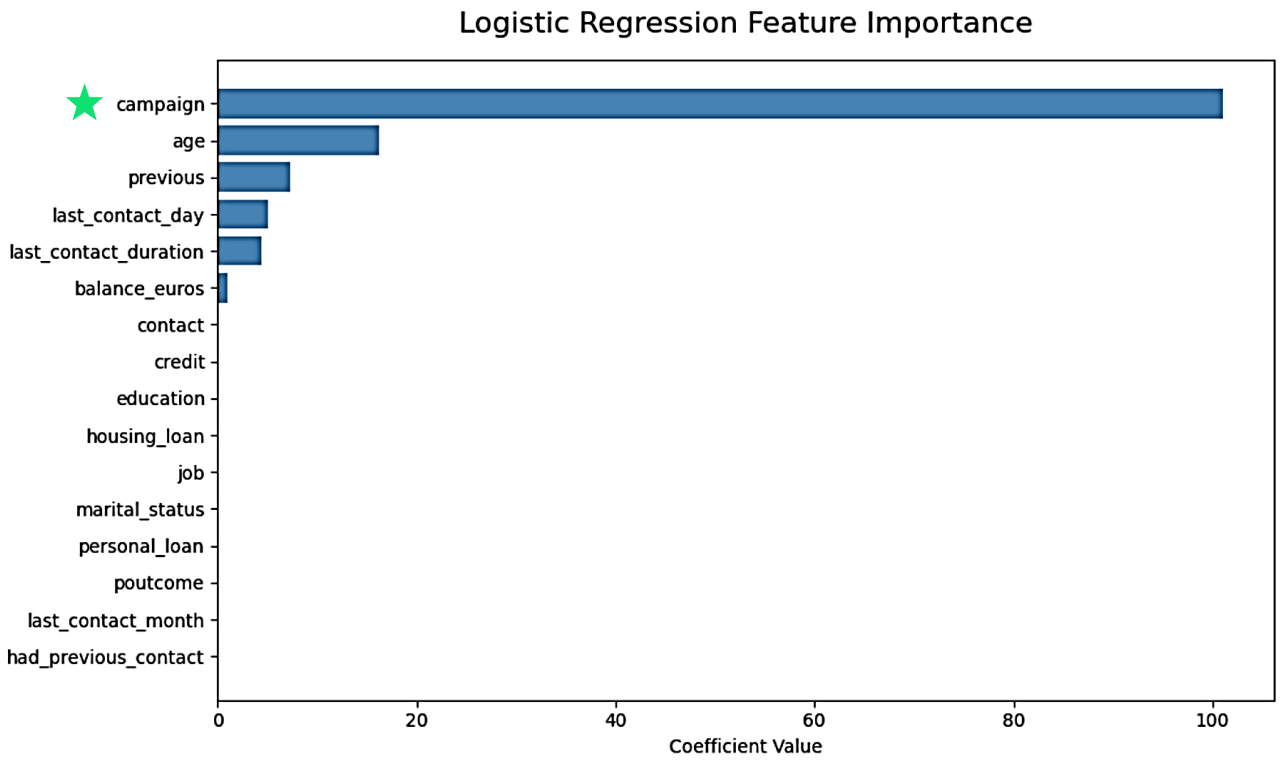
EVALUATION
ROC Curve
F1-Score
Confusion Matrix
Feature Importance

DEPLOYMENT
Logistic Regression
Test (F1-Score)
0.874

02. MACHINE LEARNING

WHAT SPARK ENABLED IN OUR ML PROCESS?

- ✓ *Spark*, alongside with *MongoDB*, allowed us to work with the entire dataset **smoothly and efficiently**, without worrying about memory or performance limitations.
- ✓ It enabled us to build a **fully automated ML pipeline**, where all preprocessing steps were handled consistently.
- ✓ We were able to train and compare **multiple models quickly**, benefiting from Spark's parallel processing and distributed architecture.
- ✓ The workflow became **reliable and reproducible**, making it easy to retrain or update models as new data becomes available.



Overall, *Spark* gave us a real **big data experience**, where the platform manages optimization and scalability, allowing us to focus on analysis and decision-making.

03. GRAPH FRAMES

BOOKS DATASET

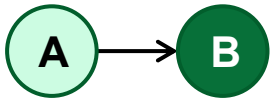
SOURCE: Goodreads-derived structured data on books, authors, genres, ratings, and reviews.

https://huggingface.co/datasets/BrightData/Goodreads-Books

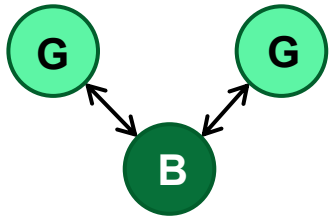
GOAL: Transform interaction logs into a Graph to build a Recommendation System and identify user communities.

Cleaned and filtered to around **82 700 rows**

AUTHORS WITH BOOKS



GENRES WITH BOOKS



BOOK INFO

- # ID
- # Kindle Price
- Name
- Genres
- # Star Rating

AUTHOR INFO

- Author Name
- # Number of books written
- # Number of follower

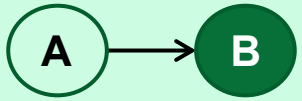
COMMUNITY REVIEWS INFO

- Num of reviews with each star rating
- # % of reviews with each star rating

03. GRAPH FRAMES

BOOKS DATASET

AUTHORS WITH BOOKS



OUTDEGREE

TOP 5 AUTHORS WITH MOST BOOKS

```
+--------------------+---------+
|id                  |outDegree|
+--------------------+---------+
|Brian Michael Bendis|59       |
|Peter David         |45       |
|Garth Ennis         |44       |
|Francine Pascal     |43       |
|Ann M. Martin       |41       |
+--------------------+---------+
only showing top 5 rows
```

MOTIF

FIVE STARS BOOKS

```
+---------------+--------------------+-----------+
|          label|               label|star_rating|
+---------------+--------------------+-----------+
|  A.K.  Agarwal|   Clinical Medicine|          5|
|  Adam Ferguson|The Correspondenc...|          5|
|      Adbusters|The Big Ideas of ...|          5|
|Adept Summaries|Summary: To Sell ...|          5|
| Aislinn Kearns|Exclusively You: ...|          5|
+---------------+--------------------+-----------+
only showing top 5 rows
```

03. GRAPH FRAMES

BOOKS DATASET

PAGE RANK
GENRE IMPORTANCE

```
+----------+-----+------------------+
|id        |type |pagerank          |
+----------+-----+------------------+
|Nonfiction|genre|3082.9607519018778|
|Romance   |genre|1864.6296883374414|
|Fiction   |genre|1607.612423343485 |
|History   |genre|1361.585582658973 |
|Poetry    |genre|959.5612467335164 |
+----------+-----+------------------+
only showing top 5 rows
```

INDEGREE
GENRE POPULARITY

```
--- Top Genres by Book Count
+----------+--------+
|id        |inDegree|
+----------+--------+
|Nonfiction|18401   |
|Fiction   |14847   |
|Romance   |10692   |
|History   |7883    |
|Fantasy   |7703    |
+----------+--------+
```

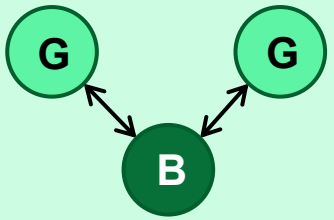
SHORTEST PATH
MINIMUM DISTANCE BETWEEN VERTICES

```
--- Shortest Path from Ethics and the Arts to Fierce Light ---

+-------------------------+-----------------------------------------------+-------------------+-------------------------------------------------------------------------------------------------+
|from                     |e0                                             |v1                 |e1                                                                                               |
+-------------------------+-----------------------------------------------+-------------------+-------------------------------------------------------------------------------------------------+
|{Ethics and the Arts, book}|{Ethics and the Arts, Philosophy, BELONGS_TO}|{Philosophy, genre}|{Philosophy, The Portable Petrarch, BELONGS_TO}                                                  |
|{Ethics and the Arts, book}|{Ethics and the Arts, Philosophy, BELONGS_TO}|{Philosophy, genre}|{Philosophy, The Buddha before Buddhism: Wisdom from the Early Teachings, BELONGS_TO}             |
|{Ethics and the Arts, book}|{Ethics and the Arts, Philosophy, BELONGS_TO}|{Philosophy, genre}|{Philosophy, Koerikoeloem, BELONGS_TO}                                                           |
|{Ethics and the Arts, book}|{Ethics and the Arts, Philosophy, BELONGS_TO}|{Philosophy, genre}|{Philosophy, The Rag and Bone Shop of the Heart: A Poetry Anthology, BELONGS_TO}                  |
|{Ethics and the Arts, book}|{Ethics and the Arts, Philosophy, BELONGS_TO}|{Philosophy, genre}|{Philosophy, The Holy Vedas ; Rig Veda, Yajur Veda, Sama Veda, Atharva Veda, BELONGS_TO}|
+-------------------------+-----------------------------------------------+-------------------+-------------------------------------------------------------------------------------------------+
```

```
+-----------------------------------------------------------+-----------------------------------------------------------------------------------------------+-----------+-----------------------------+-----------------------------------------------------------+
|v2                                                         |e2                                                                                             |v3         |e3                           |to                                                         |
+-----------------------------------------------------------+-----------------------------------------------------------------------------------------------+-----------+-----------------------------+-----------------------------------------------------------+
|{The Portable Petrarch, book}                              |{The Portable Petrarch, Poetry, BELONGS_TO}                                                    |{Poetry, genre}|{Poetry, Fierce Light, BELONGS_TO}|{Fierce Light, book}|
|{The Buddha before Buddhism: Wisdom from the Early Teachings, book}|{The Buddha before Buddhism: Wisdom from the Early Teachings, Poetry, BELONGS_TO}|{Poetry, genre}|{Poetry, Fierce Light, BELONGS_TO}|{Fierce Light, book}|
|{Koerikoeloem, book}                                       |{Koerikoeloem, Poetry, BELONGS_TO}                                                             |{Poetry, genre}|{Poetry, Fierce Light, BELONGS_TO}|{Fierce Light, book}|
|{The Rag and Bone Shop of the Heart: A Poetry Anthology, book}|{The Rag and Bone Shop of the Heart: A Poetry Anthology, Poetry, BELONGS_TO}|{Poetry, genre}|{Poetry, Fierce Light, BELONGS_TO}|{Fierce Light, book}|
|{The Holy Vedas ; Rig Veda, Yajur Veda, Sama Veda, Atharva Veda, book}|{The Holy Vedas ; Rig Veda, Yajur Veda, Sama Veda, Atharva Veda, Poetry, BELONGS_TO}|{Poetry, genre}|{Poetry, Fierce Light, BELONGS_TO}|{Fierce Light, book}|
+-----------------------------------------------------------+-----------------------------------------------------------------------------------------------+-----------+-----------------------------+-----------------------------------------------------------+
```

GENRES WITH BOOKS



CONNECTED COMPONENTS
ISOLATED BOOKS/GENRES' VERTICES

```
--- Component Sizes ---
+----------+-----+
| component|count|
+----------+-----+
|         0|88269|
|8589939010|    3|
|       519|    2|
|8589936864|    2|
|8589939282|    2|
+----------+-----+
```

```
--- Genres NOT related to
    the main library  ---
+-------------+----------+
|id           |component |
+-------------+----------+
|Androgyne    |519       |
|Numismatics  |8589939010|
|Fire Services|8589936864|
|Physiotherapy|8589939282|
+-------------+----------+
```

03. GRAPH FRAMES

BOOKS DATASET

Traditional computational tools often encounter **bottlenecks** when processing the massive edge-lists found in real-world networks. By leveraging ***Apache Spark***, we achieved **high-speed distributed graph computation** capable of scaling to **millions of entities** (books). This architecture enables the analysis of the complete dataset, eliminating the need for representative sampling.

To address the challenges inherent in iterative graph algorithms, we implemented the following optimizations:

Caching / Persisting

↓

Keeps the graph in RAM to accelerate repetitive loops (essential for *PageRank*)

Checkpointing

↓

Truncates the lineage graph to prevent *Stack Overflow* errors during deep traversals

Partitioning

↓

Optimizes data locality to reduce expensive network shuffling during joins